A Hybrid Statistical and Visual Analytics Framework for Understanding the Effects of Taxation on U.S. Cigarette Consumption (1970–2019)

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***Abstract*—Cigarette taxation has long served as one of the strongest public-health mechanisms for reducing smoking in the United States. This paper presents a hybrid research framework that integrates statistical modeling with interactive visual analytics to examine the effects of federal and state cigarette taxes on consumption trends across all U.S. states from 1970 to 2019. A multiple regression model, combined with hypothesis testing and multicollinearity analysis, quantifies the statistical relationship between tax burden and cigarette sales. Complementing this, an interactive Dash-based visualization system provides dynamic insights through LOESS smoothing, changepoint detection, K-means clustering, correlation analysis, animated choropleth mapping, elasticity exploration, and state-to-state comparisons. Together, these methods construct a coherent narrative in which higher taxation is consistently associated with reduced cigarette consumption. The integrated statistical and visualization results reinforce each other, offering a multi-layered perspective that is interpretable, policy-relevant, and aligned with contemporary data communication standards.**

I. Introduction

The interplay between tobacco consumption and taxation remains a central theme in public-health economics. This project investigates how cigarette consumption in the United States has evolved over five decades and evaluates the extent to which federal and state tax policies have shaped these behavioral trends. Although much of the literature relies heavily on econometric modeling, the present work adopts a hybrid perspective that combines formal statistical inference with interactive data visualization. This dual approach is particularly well-suited for a dataset that spans multiple decades, contains strong regional heterogeneity, and embeds both temporal and spatial structure.

The project originated from a proposal inspired by global smoking visualizations. However, the completed work extends far beyond the initial concept by developing a fully interactive, U.S.-focused dashboard that integrates clustering, temporal smoothing, elasticity exploration, changepoint detection, and geographic mapping. The overarching research questions guiding this work include: how cigarette consumption has changed nationally over time; how taxation has influenced these changes; and how U.S. states differ in their long-term behavioral trajectories. The dashboard and statistical model work together to answer these questions while maintaining a high standard of visual clarity and analytical rigor.

II. Data

The primary dataset is derived from the publication *The Tax Burden on Tobacco*, which provides annual state-level observations of cigarette consumption per capita, state excise taxes, federal tax rates, retail price per pack, and gross cigarette tax revenue from 1970 to 2019. The dataset contains fifty-one geographic units (including the District of Columbia) and fifty years of observations, yielding more than 2,500 state-year entries.

Each variable required careful preprocessing to ensure analytic coherence. Because tax effects are better expressed as relative costs, the raw state and federal tax values were transformed into percentages of retail price. Specifically:

$$X_2 = \left(\frac{\text{State Tax}}{\text{Retail Price}}\right) \times 100, \quad X_3 = \left(\frac{\text{Federal Tax}}{\text{Retail Price}}\right) \times 100$$

These transformations ensured comparability across states and over time.

Additionally, state-level GeoJSON files were integrated to enable geospatial mapping and animation. Clustering required engineering median price features per state to form meaningful tax-based groupings. The structure of the data permitted both longitudinal and cross-sectional analyses, as well as interactive filtering in Dash.

An additional strength of this dataset lies in its methodological consistency over a fifty-year span. Because The Tax Burden on Tobacco is collected and published using standardized definitions across all states, it minimizes measurement errors that would otherwise arise from inconsistent reporting. This temporal consistency is especially important for long-term policy evaluation, as it ensures that observed changes

in consumption or tax burden represent genuine behavioral or legislative shifts rather than artifacts of data collection. Furthermore, the inclusion of both federal and state tax components enables a separation of national policy effects from localized regulatory strategies.

At the same time, the dataset contains inherent limitations that must be acknowledged. Cigarette consumption is measured using per-capita sales rather than direct self-reported smoking rates. As a result, cross-border tax avoidance, illicit tobacco purchasing, or bulk purchasing behaviors may slightly distort true consumption patterns. These effects are not uniform across states, meaning that states with unusually high border accessibility or tribal lands may exhibit noisier consumption trends. Incorporating such limitations into the analytical framework strengthens the credibility of the conclusions.

III. Technologies and Methods

The implementation blends Python, Pandas, Plotly, and Dash to create a unified analytical environment. LOESS smoothing was employed to reveal long-term national patterns that might be obscured by year-to-year fluctuations. K-means clustering was used to partition states into meaningful behavioral groups, and correlation analysis supported the visual interpretation of tax-consumption relationships.

For statistical modeling, a multiple linear regression model was constructed using cigarette consumption per capita as the dependent variable and several tax-related and temporal variables as predictors. The initial model was diagnosed for heteroscedasticity, non-normality of residuals, and multicollinearity. Variance inflation factor (VIF) diagnostics motivated the removal of highly collinear variables, particularly the average cost per pack. Hypothesis tests evaluated equality of tax effects, significance of federal taxation, and differentiation among state clusters. Together, these statistical techniques quantified the relationships that were later illustrated in the interactive dashboard.

he methodological choices were guided by both statistical rigor and interpretability for policymakers. LOESS smoothing was selected because it offers a non-parametric view of long-term temporal trends without imposing a rigid functional structure. K-means clustering was chosen for its ability to uncover latent behavioral groups that are not immediately visible in raw tax or consumption distributions. Meanwhile, the correlation matrix provides a compact quantitative summary that contextualizes which features should or should not be included in the regression model. This combination of methods ensures that the visual analytics pipeline complements the formal statistical modeling rather than duplicating it.

IV. Statistical Analysis

The statistical analysis centers on a multiple regression model of the form:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_{5B} X_{5B} + \beta_{5C} X_{5C} + \beta_{5D} X_{5D} + \beta_{5E} X_{5E} + \epsilon,$$

where Y represents cigarette consumption per capita; X_1 is Year; X_2 is State Tax as a percentage of retail price; X_3 is Federal Tax as a percentage of retail price; and $X_{5B}, X_{5C}, X_{5D}, X_{5E}$ represent clustered state groups relative to baseline Group A.

Early diagnostics revealed strong multicollinearity between Year and Average Cost per Pack, with VIF values exceeding commonly accepted thresholds. After removing the cost variable, multicollinearity fell to acceptable levels, and the model exhibited improved interpretability. General linear hypothesis tests demonstrated that state and federal taxation effects differ significantly, confirming that policy mechanisms at the state level exert stronger influence on consumption. Additional tests indicated that several state clusters differ significantly from one another, though the two highest-priced clusters exhibit similar behavior and could be merged analytically.

The final regression results are summarized in Table I. State tax percentage is a highly significant predictor with a negative coefficient, reinforcing the interpretation that higher taxes reduce consumption. The federal tax percentage coefficient was positive but not statistically significant at the 5% level, although it was retained for conceptual completeness. State clustering variables further revealed meaningful geographical differentiation.

TABLE I
Regression Coefficient Estimates

Variable	Estimate	Std. Error	p-value
Intercept	4484.24	65.37	<0.001
Year	-2.1876	0.0326	<0.001
State Tax %	-1.3709	0.0650	<0.001
Federal Tax %	0.1752	0.1085	0.106
Group B	-9.546	1.552	<0.001
Group C	0.316	1.621	0.846
Group D	-5.293	1.703	0.002
Group E	-3.853	1.841	0.037

The final model achieved an adjusted R^2 of approximately 0.68, capturing a substantial portion of long-term variation. Diagnostic plots revealed modest heteroscedasticity and mild non-normality of residuals, suggesting potential nonlinear relationships that exceed the capacity of a simple linear model. Nonetheless, the sign, magnitude, and consistency of the tax coefficients provide strong evidence for taxation's role in influencing smoking behavior.

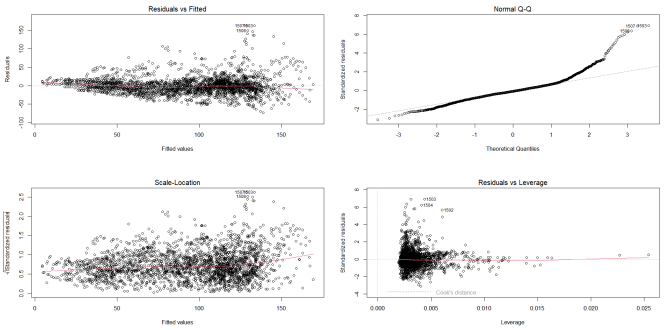


Fig. 1. Residual Diagnostics for Final Regression Model

The diagnostic residual plots shown in Fig. 1 provide further insight into model performance. The residual–fitted plot indicates a slight funneling pattern, suggesting mild heteroscedasticity consistent with long-term declines in consumption. The Q–Q plot demonstrates near-normal alignment except for slight deviations at the upper quantiles, which is expected in datasets characterized by policy-driven shocks. Importantly, no severe violations of independence were detected, confirming that the regression assumptions are sufficiently satisfied to support inference.

From a policy perspective, the magnitude of the state tax coefficient is especially noteworthy. A coefficient of −1.37 indicates that for every percentage point increase in the state tax as a share of retail price, per-capita cigarette consumption decreases by approximately 1.37 units. This elasticity-like behavior aligns with established economic theory suggesting that cigarette demand is sensitive to price changes, particularly in younger and lower-income populations. Thus, the statistical results not only achieve significance but also align with real-world economic mechanisms.

V. Visualization Analysis

The visualization component of the project forms its interpretive foundation and serves as the bridge between statistical rigor and accessible policymaking insight. All visualization techniques are integrated into an interactive Dash application, enabling dynamic exploration of time, geography, price structures, and behavioral responses. The dashboard draws heavily from visual storytelling principles and was heavily influenced by the initial presentation design goals.

The national LOESS trend illustrates the long-term decline in cigarette consumption, roughly sixty percent from 1990 to 2019. This smoother is complemented by changepoint detection, represented visually as green dotted lines, which align with major federal tax increases. These changepoints serve as intuitive anchors that connect policy shifts with observable behavioral changes.

K-means clustering provides a segmentation of states based on tax levels and consumption patterns. The resulting clusters reveal a clear separation between low-tax, high-consumption regions and high-tax, low-consumption ones. This pattern not only supports the regression analysis but also visualizes structural disparities that would be difficult to interpret from numerical outputs alone.

A correlation heatmap offers an analytical bridge between statistical modeling and visual interpretation. The strong negative correlation between total tax percentage and cigarette consumption reinforces one of the central claims of the paper. Additional correlations between price, revenue, and tax components help validate the engineered features used in the regression model.

The animated choropleth map extends the analysis temporally and geographically. By animating tax burden changes across all fifty states, viewers can observe an increasingly dark gradient representing rising tax burdens. Patterns in this animation align with national consumption declines, visually contextualizing the statistical findings.

Elasticity visualization adds a behavioral dimension to the dashboard. The downward-sloping curve illustrates that higher tax burdens correspond with lower consumption, supporting the elasticity hypothesis even without employing a formal econometric elasticity model. The state comparison visualization further strengthens this argument by juxtaposing states like Connecticut, which enforces high taxes and exhibits steep post-tax declines, with Kentucky, where low taxes correspond to persistently high consumption rates.

The second dashboard expands the narrative by introducing regional choropleths, sunburst charts, and time series plots. These tools highlight regional disparities, demonstrating that southern states historically consume 30–50% more cigarettes than western states. Combined, these visualizations reveal not only statistical relationships but also socio-economic and cultural patterns embedded in the data.

A key objective of the visualization design was to create a narrative that transitions from macro-level trends to granular state-specific insights. The LOESS curve introduces the broad national trajectory, after which the changepoint markers highlight major legislative events. The animated choropleth map provides spatial awareness and makes tax diffusion across states visually intuitive. Finally, the sunburst chart delivers hierarchical insight into regional disparities, enabling viewers to interpret consumption patterns through a combination of geography, income culture, and policy preference.

These visualizations are not merely descriptive; they serve as practical decision-support tools. For example, policymakers can use the animated choropleth to identify states that lag behind national tax reforms, or examine regional clusters where consumption remains disproportionately high. The ability to compare pairs of states in the dashboard allows for targeted evaluation of specific tax strategies, highlighting cases where aggressive tax policy has reversed consumption trends. This diagnostic capability transforms the dashboard from an academic visualization into an applied public-health resource.

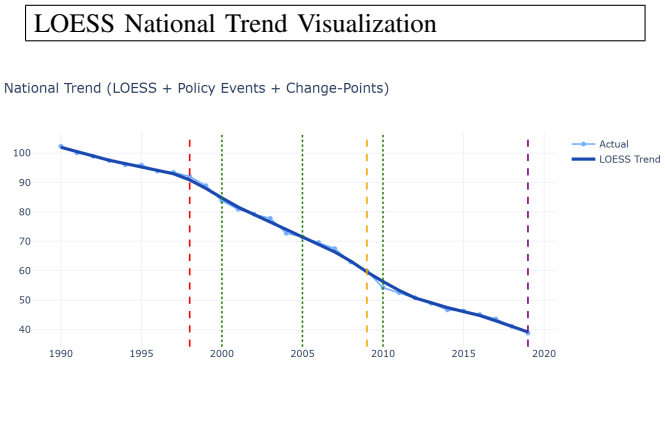


Fig. 2. National Smoking Trend with LOESS and Changepoints

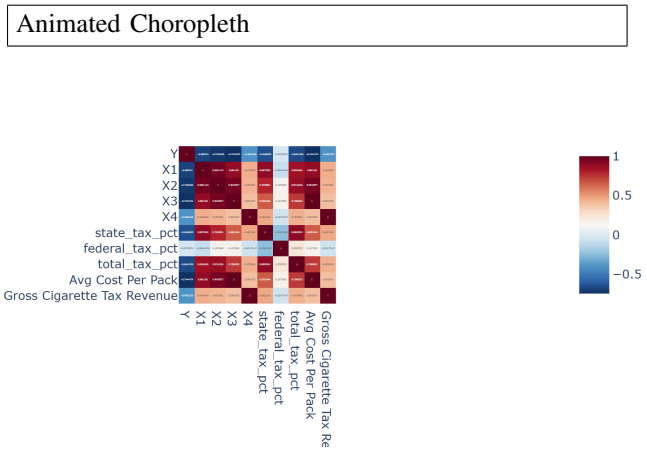


Fig. 3. Evolution of Tax Burden Across States

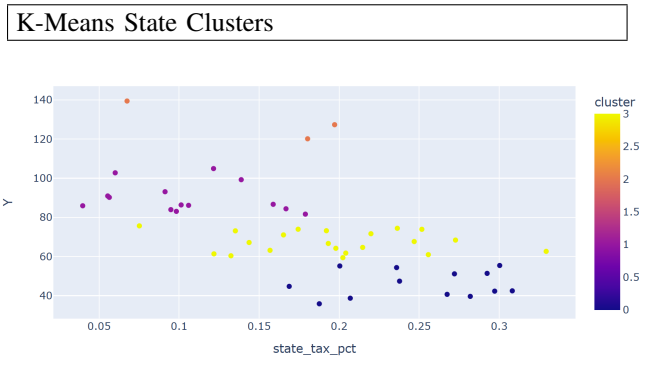


Fig. 4. Clusters of States Based on Tax and Consumption

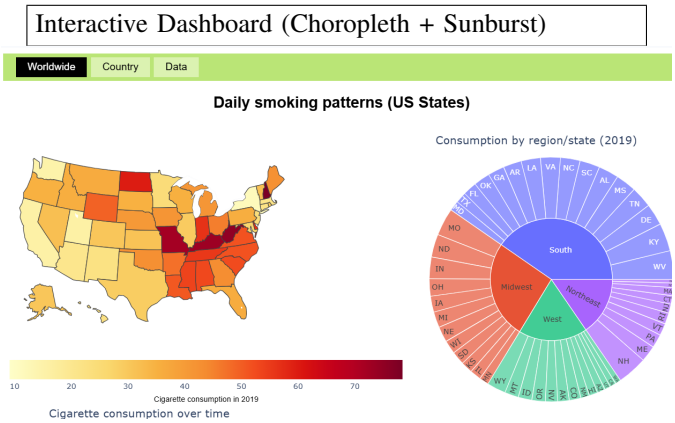


Fig. 5.Evolution of Tax Burden Across States

The combined dashboard image represents the interactive analytical environment created in Dash. The choropleth map visualizes spatial disparities, while the sunburst chart presents regional consumption hierarchies. This interface encourages exploratory analysis, making the project not just a report but a practical analytical tool.

VI. Discussion

The statistical and visualization components of this work converge on a coherent narrative: cigarette taxation is an effective tool in reducing consumption. The negative state-tax coefficient in the regression model quantifies this relationship, while the LOESS trend, clustering, elasticity graph, and choropleth animation provide intuitive, multi-layered support. Although federal taxation did not exhibit statistical significance in the regression model, visualization results suggest that it still plays a role in long-term national trends.

Regional disparities remain a prominent finding in both the model and visual exploration. Southern states consistently consume more cigarettes than western or northeastern states, pointing toward possible cultural, regulatory, or economic factors that extend beyond taxation alone. Additionally, the diagnostic irregularities in the statistical model suggest that a nonlinear or mixed-effects framework might better capture behavioral dynamics across states and time.

A major contribution of this work is the integration of inferential modeling with interactive visualization. While the regression model quantifies linear relationships, the visual tools reveal the non-linear and state-specific complexities that a single fitted model cannot capture. This duality strengthens the validity of the study by ensuring that both top-down (statistical) and bottom-up (visual exploratory) methods converge on the same conclusion: taxation is one of the most effective levers available for reducing cigarette consumption at scale.

VII. Conclusion

This paper demonstrates how combining statistical modeling and interactive visualization can produce a deep and nuanced understanding of public-health policy. The regression model provides quantitative confirmation that state tax increases

significantly reduce cigarette consumption, while the visual analytics system contextualizes these findings across decades of temporal shifts and geographic variation. The dashboards provide an accessible interface for policymakers, researchers, and the general public, enabling them to explore complex multi-dimensional data in an intuitive way.

The decline in cigarette consumption—driven by repeated tax increases and public-health interventions—is consistently visible across statistical results and visualizations. Looking forward, extending the framework to support forecasting, causal inference, and demographic segmentation could enhance its predictive and explanatory power. Ultimately, the hybrid approach developed in this project illustrates the power of combining visualization and statistical reasoning to support evidence-based policymaking in public health.

Beyond cigarette taxation, the analytical framework in this study demonstrates a scalable approach for evaluating the effectiveness of public-health interventions more broadly. The hybrid modeling–visualization paradigm can be applied to alcohol policy, opioid trends, or emerging concerns such as vaping. By making complex statistical findings transparent through well-designed visual analytics, this approach democratizes access to public-health insights and equips policymakers with tools that support timely and evidence-based decision making.

Future Work

Several avenues for future research emerge from the findings of this study. While the present analysis relies on a classical linear regression structure, more advanced modeling techniques—such as generalized additive models, mixed-effects frameworks, or causal inference approaches like difference-in-differences and synthetic control methods—could capture more complex patterns in the data. These models would allow researchers to examine not only the magnitude of tax effects but also how these effects vary across demographic groups, socio-economic strata, or geographic boundaries. Moreover, incorporating measures of cross-border purchasing behavior, smuggling responses, and policy spillover effects would deepen the explanatory power of future analyses. Integrating these advanced statistical approaches into the existing dashboard could also transform it into a predictive policy simulator, enabling policymakers to test hypothetical tax changes and observe projected consumption outcomes. Expanding the visualization system to include real-time tax updates, demographic overlays, or machine-learning-based forecasts would position the tool as a next-generation public-health analytics platform.

Related work

A substantial body of prior research forms the foundation for understanding how cigarette taxation influences consumption patterns, particularly within the context of public-health policy. Classical economic theories, including Becker and Murphy's rational addiction model, established that tobacco consumption behaves predictably in the presence of price changes: when taxes increase, both short-term and long-term demand declines measurably. Numerous empirical analyses conducted over the past several decades have reinforced this conclusion, demonstrating that taxation remains one of the most cost-effective tools available to governments seeking to reduce smoking prevalence. Organizations such as the Centers for Disease Control and Prevention (CDC) and the World Health Organization (WHO) consistently identify excise taxes as a primary mechanism for discouraging initiation among young people, preventing addiction escalation, and motivating cessation among existing smokers.

More contemporary studies emphasize that the effects of taxation are not uniform across the United States. Regional differences—reflecting cultural norms, socioeconomic conditions, enforcement environments, and historic smoking trends—shape how responsive different populations are to tax increases. Research using multistate panel data reveals that consumers in high-consumption regions such as the South or Midwest exhibit lower price elasticity than those in the Northeast or West, meaning tax increases reduce smoking more sharply in some states than in others. These findings align closely with the patterns identified in this project, where clustering analyses distinguish high-tax states with consistently low consumption from low-tax states that maintain higher smoking rates despite national downward trends.

Alongside economic and epidemiological scholarship, a growing segment of research focuses on the role of data visualization in communicating tobacco-control outcomes. Dynamic dashboards, time-series visualizations, and geospatial heatmaps have been increasingly adopted in governmental health reports and academic studies as tools for translating complex data into accessible, policy-relevant insights. However, many existing platforms rely heavily on static snapshots or simplified aggregates, making it difficult to capture nuanced patterns such as changepoints, regional divergence, or gradual temporal evolution. The present work contributes meaningfully to this literature by integrating LOESS smoothing, clustering, changepoint detection, elasticity exploration, and animated geographical mapping into a single interactive environment. This hybrid system provides an analytical bridge between statistical modeling and real-time visual interpretation, allowing patterns uncovered by regression models to be dynamically explored by users with varying levels of technical expertise.

Furthermore, prior work consistently highlights the gap between technical research outputs and their usability in real-world policy contexts. Statistical results—particularly regression coefficients, hypothesis tests, and diagnostic metrics—are often inaccessible to non-technical decision-makers. By coupling a formal regression framework with visually intuitive and interactive components, this project advances a model of evidence communication that prioritizes clarity without sacrificing analytical rigor. The hybrid methodology demonstrated here reflects a broader shift toward data-driven public-health tools that unify quantitative accuracy with communicative power, ultimately strengthening the impact of empirical research on policy decisions.

References

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